

Dominic Bunnett

dombunnett.github.io

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Research interests

Moduli theory and geometric invariant theory, at the interface of algebraic and discrete geometry, with an emphasis on computational and algorithmic problems.

Education

- **Freie Universität Berlin** October 2015 – July 2019

Doctor of Philosophy in Mathematics

Thesis: Moduli of hypersurfaces in toric orbifolds

Advisor: Prof. Victoria Hoskins

- **University of Warwick** 2011 – 2015

BSc and MMath, 1st class

Thesis: Hodge theory and the Torelli theorem for curves

Advisor: Prof. Miles Reid

Positions held

- **Technische Universität Berlin** October 2021 – present

Postdoc, research group Algorithmic Algebra

Mentor: Peter Bürgisser

- **The Open University, UK** September 2024 – present

Associate Lecturer, *Applied Computational Mathematics*

- **Makerere University, Kampala** April 2022 – January 2023

Lecturer

- **Technische Universität Berlin** August 2019 – May 2021

Postdoc within the Einstein Visiting Fellowship of Bernd Sturmfels (Einstein Foundation Berlin); initially as a Stipendiat, subsequently employed

Mentors: Michael Joswig and Bernd Sturmfels

- **University of Oxford** April – June 2019

Visiting Researcher, hosted by Frances Kirwan

- **Freie Universität Berlin** September 2015 – March 2019

Doktorand / Wissenschaftlicher Mitarbeiter

Advisor: Victoria Hoskins

Publications

- On the moduli of hypersurfaces in toric orbifolds. *Proceedings of the Edinburgh Mathematical Society*, 67(2):577–616, 2024. doi:10.1017/S0013091524000166, arXiv:1906.00272
- Generalised cone complexes and tropical moduli in polymake. *Joint with Michael Joswig and Julian Pfeifle*. In *Proceedings of the 2023 International Symposium on Symbolic and Algebraic Computation*

(ISSAC '23), pages 100–106. ACM, New York, 2023. doi:10.1145/3597066.3597093

- Determinantal representations of the cubic discriminant. *Joint with Hanieh Keneshlou. Le Matematiche*, 75(2):489–505, 2020. journal, arXiv:1909.05579

Preprints

- Finite automorphism groups of hypersurfaces via Bott vanishing. *Joint with Caroline Namanya*. arXiv:2609.38572
- Embeddings of weighted projective spaces. *Joint with Praise Adeyemo and Fabián Levicán-Santibáñez*. arXiv:2510.05076 (submitted)
- Moduli of Bridgeland semistable holomorphic triples. *Joint with Alejandra Rincón-Hidalgo*. arXiv:2102.04995 (submitted)

In preparation

- K-moduli of exceptional del Pezzo hypersurfaces. *Joint with Josh Jackson, Jesús Martínez-García and Charlotte Satchwell*.
- Geometry of moduli stacks of hypersurfaces. *Joint with Josh Jackson*.
- Hilbert–Mumford criteria for hypersurfaces.

Awards and competitive research programmes

- **Research in Teams**, Erwin Schrödinger International Institute for Mathematics and Physics (ESI), Vienna, May – June 2025.
Project: *Projective geometry, combinatorics and representation theory from polytopes*, with Praise Adeyemo, Livia Campo and Balázs Szendrői.
- **Research in Pairs**, Mathematisches Forschungsinstitut Oberwolfach, 2021.
Awarded jointly with Alejandra Rincón-Hidalgo.
- **Clay Mathematics Institute**, full funding for the workshop *Equivariant methods in geometry*, University of Ibadan, 2024.
- **MATH+ travel grants**, for research and teaching visits to Makerere University and AIMS Ghana.
- **International Centre for Theoretical Physics**, funding for research visits to Trieste (2019, 2022) and for the EAUMP–ICTP summer school, University of Dar-es-Salaam, 2018.

Invited talks

- AGGITatE 2026, workshop of the Isaac Newton Institute satellite programme *Algebraic Groups, Geometry, Invariants and Related Topics*, University of Essex, July 2026.
Finite automorphisms of toric orbifold hypersurfaces.
- *Embeddings of weighted projective spaces and rectangular simplices*. Discrete Mathematics / Geometry seminar, TU Berlin, November 2025.
- *Embedding weighted projective spaces*. Workshop in Algebraic Geometry, Erwin Schrödinger International Institute for Mathematics and Physics, Vienna, 2025.
- *Families of hypersurfaces from polytopes*. Conference *Algebra and Geometry from Africa*, ICMS, Edinburgh, May 2025.
- *The Frobenius problem, simplices and moduli spaces*. Mathematics and Statistics Colloquium, The Open University, February 2025.
- *Moduli of coverings branched along a hypersurface*. Programme seminar, *New Equivariant Methods in*

Algebraic and Differential Geometry, Isaac Newton Institute, Cambridge, July 2024.

- *Building collaborations with Africa*. Symposium *Building partnerships with institutions and researchers in Africa*, Institute for Advanced Study, University of Amsterdam, June 2024.
- *Geometry of some moduli spaces of hypersurfaces*. COW Seminar, University of Bath, March 2024.
- *Moduli spaces of hypersurfaces and their topology*. Mathematics Essex Seminar Series, University of Essex, December 2023.
- *Stacky fans and tropical moduli in polymake*. ISSAC 2023, Tromsø, Norway, July 2023.
- *Moduli spaces of weighted hypersurfaces*. Geometry Seminar, Radboud University Nijmegen, March 2023.
- *Moduli of Bridgeland stable triples*. International Centre for Theoretical Physics, Trieste, March 2022.
- *Projective moduli of semistable hypersurfaces*. Geometry and Topology Seminar, Universität Wien, January 2023.
- *Normal simplices and moduli of weighted hypersurfaces*. African Mathematics Seminar, April 2021.
- *polymake introduction*. 11th polymake Conference and Developer Meeting, MPI for Mathematics in the Sciences, Leipzig, January 2020.
- *What are algebraic curves and their tropical friends?* “What is...?” Seminar, Berlin Mathematical School, Urania Berlin, November 2019. *Joint with Marta Panizzut*.
- *What is $M_{g,n}$?* “What is...?” Seminar, Berlin Mathematical School, FU Berlin, October 2019.
- *On the moduli of weighted hypersurfaces*. Max Planck Institute for Mathematics in the Sciences, Leipzig, 2019.
- *Moduli of hypersurfaces in weighted projective space*. Algebraic Geometry Seminar, University of Oxford, May 2019.
- *Moduli of hypersurfaces in weighted projective space*. International Centre for Theoretical Physics, Trieste, January 2019.
- *On the moduli of weighted hypersurfaces*. EAUMP–ICTP summer school on homological methods in algebra and geometry, University of Dar-es-Salaam, July 2018.
- *Quotients in classification problems*. American University in Cairo, September 2017.

Supervision

- David Schmalzing, Bachelor’s thesis, *Regular sequences and stabiliser groups of polynomials*, TU Berlin, 2025
- Josef Cutler, Bachelor’s thesis, *Locally nilpotent derivations and $\mathbb{G}_a$ -actions*, TU Berlin, 2025
- Yosime Flood, Bachelor’s thesis, *Algebraic curves and the Riemann–Roch theorem*, TU Berlin, 2024
- Jonathan Wasserman, Master’s thesis, *The Keel–Mori theorem*, TU Berlin, in progress
- Aleksandra Kozhevnikova, Master’s thesis, *Borel–Moore cohomology of geometric quotients*, TU Berlin, in progress
- Two Bachelor’s theses, TU Berlin, in progress

Mathematics in Africa

Sustained involvement since 2014, in teaching, organisation, programme governance and outreach. Individual entries also appear under Teaching, Organisational work and Research visits.

- Lecturer, Makerere University, Kampala, 2022–23; guest lecturer for a graduate seminar there, 2023; organiser of a workshop there, 2021, and co-organiser of a conference there, 2027 (forthcoming)
- Programme committee, Young African Mathematicians fellowship programme, MATH+, Berlin; international collaborator, Eastern Africa Algebra Research Group

- Organiser and mini-course lecturer, workshop *Equivariant methods in geometry*, University of Ibadan, 2024
- Lecture course on lattice polytopes, *Algebra and Geometry from Africa* (Mathematics for Humanity programme), ICMS Edinburgh, 2025
- Guest lecturer, AIMS Ghana, 2020 and 2021/22; research visit to AIMS Rwanda, 2023
- Facilitator and speaker, EAUMP–ICTP summer school, University of Dar-es-Salaam, 2018; lecturer, 1st EAALG workshop, 2021
- Public talks *Why mathematics?* in Kampala, 2022; A-level teaching and university-application mentoring, St. Peter’s Kubatana High School, Harare, Zimbabwe, 2014

Service and outreach

- Nominated and prepared the successful application for a MATH+ Hanna Neumann Fellowship (Caroline Namanya, 2026)
- Referee for *Bulletin of the London Mathematical Society*, *Journal of the London Mathematical Society*, *International Mathematics Research Notices* and *Advances in Geometry*
- Programme committee, Young African Mathematicians (YAM) fellowship programme, MATH+, Berlin
- International collaborator, Eastern Africa Algebra Research Group (EAALG)
- *Calculate with Africa: transforming the world through mathematics*, Meridian podcast, Berlin University Alliance, 2024
- *Why mathematics?* Public talks for incoming university students at Makerere University and for pupils at a state secondary school in Kampala, 2022

Research visits

- Isaac Newton Institute satellite programme *Algebraic Groups, Geometry, Invariants and Related Topics*, University of Essex, invited participant, July 2026
- Isaac Newton Institute, University of Cambridge, programme *New Equivariant Methods in Algebraic and Differential Geometry*, January – June 2024
- AIMS Rwanda, *visiting* Jan Hązła, January 2023
- University of Essex, Colchester, *visiting* Jesús Martínez-García, April 2022
- International Centre for Theoretical Physics, Trieste, *visiting* Alejandra Rincón, March 2022
- University of Oxford, *visiting* Praise Adeyemo and Balázs Szendrői, January 2022
- International Centre for Theoretical Physics, Trieste, *visiting* Tarig Abdelgadir, January 2019

Software

Available at <https://github.com/dombunnett>

- Stacky fans and tropical moduli in polymake, `polymake`
Joint with Michael Joswig and Julian Pfeifle.
- Reduced submodules of finite dimensional polynomial modules, `python`
- Determinantal representations of the cubic discriminant, `M2`
- Automorphism group of a projective toric orbifold, `polymake`
- Non-normal simplices and weighted projective spaces, `polymake`

Organisational work

- Co-organiser: Conference in algebraic geometry, Makerere University, Kampala, 2027 (forthcoming)
- Organiser: Workshop *Equivariant methods in geometry*, University of Ibadan, January 2024
- Organiser: Workshop on homological algebra and discrete geometry, Makerere University, Kampala, 2021
- Organiser: Workshop on nonlinear algebra, TU Berlin, 2020
- Group leader: Cubic surfaces workshop, University of Oslo, 2019
- Facilitator: EAUMP–ICTP summer school on homological methods in algebra and geometry, University of Dar-es-Salaam, 2018
- Organiser: Research seminar — further topics in geometric invariant theory, Freie Universität Berlin, Summer Semester 2016

Teaching

- Summer Semester 2026, Algebra II (Commutative Algebra), TU Berlin
- Summer Semester 2026, Seminar on Linear Algebraic Groups, TU Berlin
- Winter Semester 2025/26, Algebraic Geometry I, TU Berlin
- Winter Semester 2025/26, Analysis II for Engineers, TU Berlin
- Since September 2024, *Applied Computational Mathematics*, The Open University (Associate Lecturer)
- May 2025, Lecture course on lattice polytopes, *Algebra and Geometry from Africa*, ICMS, Edinburgh
- Summer Semester 2025, Schemes reading course, TU Berlin
- Summer Semester 2024, Algebra II (Commutative Algebra), TU Berlin
- Winter Semester 2023/24, Algebra I (Fields and Galois Theory), TU Berlin
- January 2024, Mini-course on moduli spaces and group actions, workshop *Equivariant methods in geometry*, University of Ibadan
- Winter Semester 2023/24, Algebraic Geometry (remote learning course)
- August 2023, Graduate seminar on advanced topics in commutative algebra and geometry, Makerere University
- Summer Semester 2023, Calculus I and Linear Algebra for Engineers, TU Berlin
- Summer Semester 2022, Algebraic Geometry, Makerere University
- Winter Semester 2021/22, Algebra I (Galois Theory, auf Deutsch), TU Berlin
- Winter Semester 2021/22, Computational introduction to algebraic geometry, AIMS Ghana
- December 2021, Introduction to homological algebra, EAALG workshop
- Summer Semester 2020, Derived categories in algebraic geometry, Makerere University
- 2020, Polyhedral geometry and polymake, Abram Gannibal workshop
- 2020, Singularities of affine hypersurfaces, Macaulay2 workshop
- Winter Semester 2017/18, Linear Algebra I (auf Deutsch), FU Berlin
- Summer Semester 2017, Linear Algebra II (auf Deutsch), FU Berlin
- Winter Semester 2016/17, Differential Geometry I, Berlin Mathematical School
- 2014–2015, Undergraduate tutor (analysis, linear algebra, abstract algebra, calculus), University of Warwick; 2012–2015, private tutor at A-level and GCSE; Summer 2014, A-level mathematics, St. Peter's Kubatana High School, Harare, Zimbabwe

Languages

English (native) · German (fluent) · Luganda (beginner)